

B3: Quantum, Atomic and Molecular Physics — Model Solutions, 2018

Question 1 — Fine structure of hydrogenic systems

(a) Origin of fine structure and derivation of ΔE_{so}

Problem. Explain the physical origin of fine structure in hydrogenic atoms; derive $\Delta E_{\text{so}} = \frac{\beta}{2}[j(j+1) - l(l+1) - s(s+1)]$ and find β .

Physical origin: in the electron rest frame the proton orbits, generating a magnetic field $\mathbf{B}$; the electron spin magnetic moment $\boldsymbol{\mu}_s = -g_s \mu_B \mathbf{S}/\hbar$ couples to $\mathbf{B}$, giving the spin-orbit interaction (a relativistic correction of order α^2).

Spin-orbit Hamiltonian (Thomas-corrected):

$$H_{\text{so}} = \frac{1}{2m_e^2 c^2} \frac{1}{r} \frac{dV}{dr} \mathbf{L} \cdot \mathbf{S}, \quad V(r) = -\frac{Ze^2}{4\pi\epsilon_0 r}.$$

Differentiate V :

$$\frac{dV}{dr} = \frac{Ze^2}{4\pi\epsilon_0} \frac{1}{r^2}.$$

Divide by r :

$$\frac{1}{r} \frac{dV}{dr} = \frac{Ze^2}{4\pi\epsilon_0} \frac{1}{r^3}.$$

Define $\xi(r) = (2m_e^2 c^2)^{-1}(1/r)(dV/dr)$:

$$H_{\text{so}} = \xi(r) \mathbf{L} \cdot \mathbf{S}.$$

Use $\mathbf{J} = \mathbf{L} + \mathbf{S}$:

$$\mathbf{J}^2 = \mathbf{L}^2 + \mathbf{S}^2 + 2\mathbf{L} \cdot \mathbf{S}.$$

Solve for $\mathbf{L} \cdot \mathbf{S}$:

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2}(\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2).$$

Take eigenvalues in $|n l s j m_j\rangle$:

$$\langle \mathbf{L} \cdot \mathbf{S} \rangle = \frac{\hbar^2}{2} [j(j+1) - l(l+1) - s(s+1)].$$

Take the radial expectation:

$$\Delta E_{\text{so}} = \langle \xi(r) \rangle \frac{\hbar^2}{2} [j(j+1) - l(l+1) - s(s+1)].$$

Identify $\beta = \hbar^2 \langle \xi(r) \rangle$:

$$\beta = \frac{\hbar^2}{2m_e^2 c^2} \frac{Ze^2}{4\pi\epsilon_0} \left\langle \frac{1}{r^3} \right\rangle.$$

Insert the hint for $\langle 1/r^3 \rangle$:

$$\beta = \frac{\hbar^2}{2m_e^2 c^2} \frac{Ze^2}{4\pi\epsilon_0} \left(\frac{Z}{na_0} \right)^3 \frac{1}{l(l + \frac{1}{2})(l + 1)}.$$

Collect powers of Z :

$$\beta = \frac{\hbar^2 Z^4 e^2}{8\pi\epsilon_0 m_e^2 c^2 n^3 a_0^3 l(l + \frac{1}{2})(l + 1)}.$$

Substitute $a_0 = 4\pi\epsilon_0 \hbar^2 / (m_e e^2)$:

$$a_0^3 = \frac{(4\pi\epsilon_0)^3 \hbar^6}{m_e^3 e^6}.$$

Insert into β :

$$\beta = \frac{\hbar^2 Z^4 e^2}{8\pi\epsilon_0 m_e^2 c^2 n^3 l(l + \frac{1}{2})(l + 1)} \cdot \frac{m_e^3 e^6}{(4\pi\epsilon_0)^3 \hbar^6}.$$

Collect factors:

$$\beta = \frac{Z^4 m_e e^8}{2(4\pi\epsilon_0)^4 \hbar^4 c^2 n^3 l(l + \frac{1}{2})(l + 1)}.$$

Identify $\alpha = e^2 / (4\pi\epsilon_0 \hbar c)$:

$$\alpha^4 = \frac{e^8}{(4\pi\epsilon_0)^4 \hbar^4 c^4}.$$

Multiply top and bottom by c^2 :

$$m_e \cdot \frac{e^8}{(4\pi\epsilon_0)^4 \hbar^4 c^2} = m_e c^2 \cdot \frac{e^8}{(4\pi\epsilon_0)^4 \hbar^4 c^4}.$$

Identify the right-hand factor as α^4 :

$$m_e \cdot \frac{e^8}{(4\pi\epsilon_0)^4 \hbar^4 c^2} = \alpha^4 m_e c^2.$$

Substitute and simplify:

$$\boxed{\beta = \frac{Z^4 \alpha^4 m_e c^2}{2n^3 l(l + \frac{1}{2})(l + 1)}}$$

(equivalently $\beta = -E_n Z^2 \alpha^2 / [n l(l + \frac{1}{2})(l + 1)]$ with $E_n = -\frac{1}{2} m_e c^2 Z^2 \alpha^2 / n^2$).

(b) Fine structure of $n = 2, 3$ in He^+

He^+ : $Z = 2$, $s = 1/2$. Combine $\Delta E_{\text{so}} = (\beta/2)[\dots]$ with β from (a):

$$\Delta E_{\text{so}} = \frac{Z^4 m_e c^2 \alpha^4}{4n^3 l(l + \frac{1}{2})(l + 1)} [j(j + 1) - l(l + 1) - s(s + 1)].$$

Evaluate α^4 with $\alpha = 1/137.036$:

$$\alpha^4 = (1/137.036)^4 = 2.834 \times 10^{-9}.$$

Multiply by $m_e c^2 = 0.5110 \times 10^6 \text{ eV}$:

$$m_e c^2 \alpha^4 = 0.5110 \times 10^6 \text{ eV} \times 2.834 \times 10^{-9} = 1.449 \text{ meV}.$$

$n = 2$: levels $2s_{1/2}$, $2p_{1/2}$, $2p_{3/2}$.

- $2s_{1/2}$: $l = 0$ (no SO shift; included for completeness).
- $2p_{1/2}$: $l = 1$, $s = 1/2$, $j = 1/2$.

Compute the angular bracket:

$$j(j+1) - l(l+1) - s(s+1) = \frac{3}{4} - 2 - \frac{3}{4} = -2.$$

Compute the orbital denominator:

$$l(l + \frac{1}{2})(l+1) = 1 \cdot \frac{3}{2} \cdot 2 = 3.$$

Insert into ΔE_{so} with $Z^4 = 16$, $n^3 = 8$:

$$\Delta E(2p_{1/2}) = \frac{16}{4 \cdot 8 \cdot 3} \cdot (-2) \cdot 1.449 \text{ meV}.$$

Simplify the prefactor:

$$\Delta E(2p_{1/2}) = -\frac{1}{3} \cdot 1.449 \text{ meV} = -0.483 \text{ meV}.$$

- $2p_{3/2}$: $l = 1$, $s = 1/2$, $j = 3/2$.

Compute the angular bracket:

$$j(j+1) - l(l+1) - s(s+1) = \frac{15}{4} - 2 - \frac{3}{4} = +1.$$

Insert into ΔE_{so} :

$$\Delta E(2p_{3/2}) = \frac{16}{4 \cdot 8 \cdot 3} \cdot (+1) \cdot 1.449 \text{ meV} = +0.242 \text{ meV}.$$

$n = 3$: levels $3s_{1/2}$, $3p_{1/2}$, $3p_{3/2}$, $3d_{3/2}$, $3d_{5/2}$.

Compute overall prefactor for $n = 3$, $Z = 2$:

$$\frac{Z^4 m_e c^2 \alpha^4}{4n^3} = \frac{16}{4 \cdot 27} \cdot 1.449 \text{ meV}.$$

Simplify:

$$\frac{Z^4 m_e c^2 \alpha^4}{4n^3} = \frac{16}{108} \cdot 1.449 \text{ meV} = 0.2147 \text{ meV}.$$

Compute angular ratios $[j(j+1) - l(l+1) - s(s+1)]/[l(l + \frac{1}{2})(l+1)]$:

$$\begin{aligned} 3p_{1/2} : \frac{-2}{3}, \quad 3p_{3/2} : \frac{1}{3}, \\ 3d_{3/2} : \frac{-3}{15} = -\frac{1}{5}, \quad 3d_{5/2} : \frac{2}{15}. \end{aligned}$$

Multiply each ratio by 0.2147 meV :

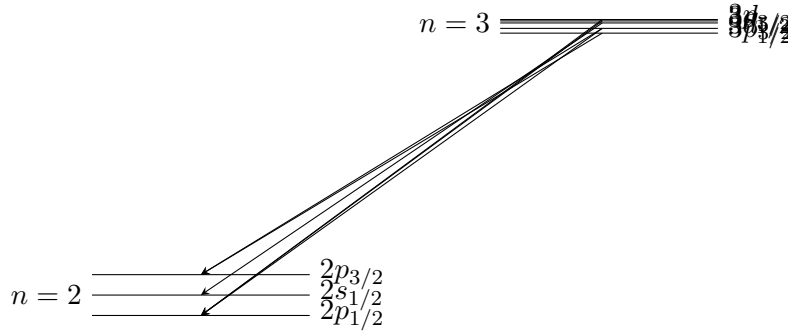
$$\Delta E(3p_{1/2}) = -\frac{2}{3} \cdot 0.2147 \text{ meV} = -0.143 \text{ meV},$$

$$\Delta E(3p_{3/2}) = +\frac{1}{3} \cdot 0.2147 \text{ meV} = +0.0716 \text{ meV},$$

$$\Delta E(3d_{3/2}) = -\frac{1}{5} \cdot 0.2147 \text{ meV} = -0.0429 \text{ meV},$$

$$\Delta E(3d_{5/2}) = +\frac{2}{15} \cdot 0.2147 \text{ meV} = +0.0286 \text{ meV}.$$

Energy-level diagram:



Allowed transitions ($\Delta l = \pm 1$, $\Delta j = 0, \pm 1$, $j = 0 \not\rightarrow 0$):

$$3p_{1/2} \rightarrow 2s_{1/2}, \quad 3p_{3/2} \rightarrow 2s_{1/2}, \quad 3s_{1/2} \rightarrow 2p_{1/2}, \quad 3s_{1/2} \rightarrow 2p_{3/2},$$

$$3d_{3/2} \rightarrow 2p_{1/2}, \quad 3d_{3/2} \rightarrow 2p_{3/2}, \quad 3d_{5/2} \rightarrow 2p_{3/2}.$$

(c) Ω^- bound to bare Pb ($Z=82$), $n = 10$, $l = 9$

Splitting into 4 levels means $2s + 1 = 4$:

$$s_{\Omega} = 3/2$$

Possible j values $j = l - s, \dots, l + s$:

$$j = 15/2, 17/2, 19/2, 21/2.$$

Reduced-mass effect: replace $m_e \rightarrow \mu = m_{\Omega} m_{\text{Pb}} / (m_{\Omega} + m_{\text{Pb}}) \approx m_{\Omega}$ since the Pb nucleus is much heavier. Apply formula from (a) with $m_e \rightarrow m_{\Omega}$:

$$\beta = \frac{Z^4 \alpha^4 m_{\Omega} c^2}{2n^3 l(l + \frac{1}{2})(l + 1)}.$$

Evaluate Z^4 :

$$Z^4 = 82^4 = 4.5212 \times 10^7.$$

Evaluate α^4 :

$$\alpha^4 = (1/137.036)^4 = 2.834 \times 10^{-9}.$$

Multiply:

$$Z^4 \alpha^4 = 0.1281.$$

Evaluate angular denominator $l(l + \frac{1}{2})(l + 1)$:

$$9 \cdot 9.5 \cdot 10 = 855.$$

Combine with $n^3 = 1000$:

$$\frac{Z^4 \alpha^4}{2n^3 l(l + \frac{1}{2})(l + 1)} = \frac{0.1281}{2 \cdot 1000 \cdot 855} = 7.49 \times 10^{-8}.$$

Multiply by $m_{\Omega} c^2 = 1.672 \times 10^9 \text{ eV}$:

$$\beta = 7.49 \times 10^{-8} \times 1.672 \times 10^9 \text{ eV} = 125.2 \text{ eV}.$$

Write the energy of level j :

$$E_j = \frac{\beta}{2} [j(j+1) - l(l+1) - s(s+1)].$$

Insert $l(l+1) = 90$ and $s(s+1) = 15/4$:

$$E_j = \frac{\beta}{2} [j(j+1) - 90 - \frac{15}{4}].$$

Take the difference $E_{j+1} - E_j$:

$$E_{j+1} - E_j = \frac{\beta}{2} [(j+1)(j+2) - j(j+1)].$$

Factor out $(j+1)$:

$$(j+1)(j+2) - j(j+1) = (j+1)[(j+2) - j] = 2(j+1).$$

Substitute back:

$$E_{j+1} - E_j = \beta(j+1).$$

Insert $j = 15/2, 17/2, 19/2$:

$$E_{17/2} - E_{15/2} = \frac{17}{2}\beta, \quad E_{19/2} - E_{17/2} = \frac{19}{2}\beta, \quad E_{21/2} - E_{19/2} = \frac{21}{2}\beta.$$

Multiply by $\beta = 125.2 \text{ eV}$:

$$\Delta E_{15/2 \rightarrow 17/2} = \frac{17}{2} \cdot 125.2 \text{ eV} = 1064 \text{ eV},$$

$$\Delta E_{17/2 \rightarrow 19/2} = \frac{19}{2} \cdot 125.2 \text{ eV} = 1189 \text{ eV},$$

$$\Delta E_{19/2 \rightarrow 21/2} = \frac{21}{2} \cdot 125.2 \text{ eV} = 1315 \text{ eV}.$$

Convert to keV:

$\Delta E_{15/2 \rightarrow 17/2} = 1.06 \text{ keV}, \quad \Delta E_{17/2 \rightarrow 19/2} = 1.19 \text{ keV}, \quad \Delta E_{19/2 \rightarrow 21/2} = 1.31 \text{ keV}.$

(d) Two Ω^- bound to Pb: configurations $1s2l$ and ground

The Ω^- has $s = 3/2$ (half-integer), so it is a fermion. Pauli exclusion demands total wavefunction antisymmetric under exchange.

Couple two spin-3/2 particles:

$$3/2 \otimes 3/2 = 3 \oplus 2 \oplus 1 \oplus 0.$$

Identify exchange symmetry (standard result for two identical spin- s):

- $S = 3, 1$: spin symmetric;
- $S = 2, 0$: spin antisymmetric.

Excited $1s2l$ configurations ($l = 0, 1$): two particles in different spatial orbitals; spatial part can be either symmetric or antisymmetric.

For $1s2s$ ($L = 0$), pair spatial $\pm$ with spin $\mp$:

$$\text{spatial sym} \otimes \text{spin antisym} : {}^1S_0, {}^5S_2,$$

$$\text{spatial antisym} \otimes \text{spin sym} : {}^3S_1, {}^7S_3.$$

Collect levels for $1s2s$:

$$\boxed{{}^1S_0, {}^3S_1, {}^5S_2, {}^7S_3}.$$

For $1s2p$ ($L = 1$), repeat the pairing:

$$\text{spatial sym} \otimes \text{spin antisym} : {}^1P_1, {}^5P_{1,2,3},$$

$$\text{spatial antisym} \otimes \text{spin sym} : {}^3P_{0,1,2}, {}^7P_{2,3,4}.$$

Collect levels for $1s2p$:

$$\boxed{{}^1P_1, {}^3P_{0,1,2}, {}^5P_{1,2,3}, {}^7P_{2,3,4}}.$$

Ground configuration $1s^2$: both particles in the same spatial orbital, so spatial part is necessarily symmetric. Spin must therefore be antisymmetric:

$$S = 0, 2 \Rightarrow \boxed{\text{ground: } {}^1S_0 \text{ and } {}^5S_2}.$$

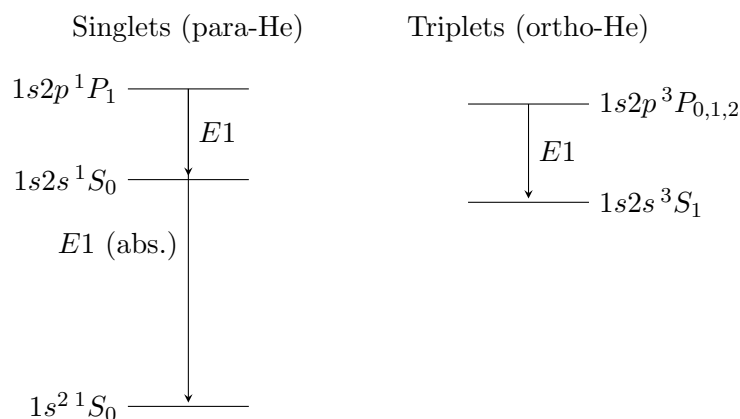
Comparison with He: two spin-1/2 electrons in $1s^2$ admit only the antisymmetric singlet $S = 0$, giving uniquely 1S_0 . Two spin-3/2 Ω^- admit *two* antisymmetric spin states ($S = 0, 2$), so the ground configuration contains both 1S_0 and 5S_2 — a richer multiplet structure than helium.

Question 2 — Helium, Zeeman effect, positronium

(a) He levels for $1s^2$, $1s2s$, $1s2p$

Configurations and terms:

- $1s^2$: 1S_0 only (Pauli; spatial sym requires spin antisym).
- $1s2s$: ${}^1S_0, {}^3S_1$.
- $1s2p$: ${}^1P_1, {}^3P_{0,1,2}$.



Selection rules ($\Delta S = 0$, $\Delta L = \pm 1$, $\Delta J = 0, \pm 1$, $J = 0 \not\rightarrow 0$, parity changes): allowed E1 transitions are $1s2p\,{}^1P_1 \rightarrow 1s^2\,{}^1S_0$, $1s2p\,{}^1P_1 \rightarrow 1s2s\,{}^1S_0$, and $1s2p\,{}^3P_{0,1,2} \rightarrow 1s2s\,{}^3S_1$.

Absorption: only from the ground state $1s^2\,{}^1S_0$, hence only $1s^2\,{}^1S_0 \rightarrow 1s2p\,{}^1P_1$.

Intercombination lines (singlet-triplet): forbidden by $\Delta S = 0$ to first order. He is light, so spin-orbit mixing is tiny; intercombination lines are very weak (the famous $1s2s\,{}^3S_1 \rightarrow 1s^2\,{}^1S_0$ is metastable). **Not** prominent.

(b) Weak-field Zeeman effect in LS coupling

Weak field: $H_{\text{Zeeman}} \ll H_{\text{so}}$, so $\mathbf{J}$ is a good quantum number; $\mathbf{L}$ and $\mathbf{S}$ precess about $\mathbf{J}$, while $\mathbf{J}$ precesses slowly about $\mathbf{B}$.

Write the Zeeman Hamiltonian:

$$H_Z = \frac{\mu_B}{\hbar} (\mathbf{L} + g_s \mathbf{S}) \cdot \mathbf{B}, \quad g_s \approx 2.$$

Take $\mathbf{B} = B\hat{z}$:

$$H_Z = \frac{\mu_B B}{\hbar} (L_z + 2S_z).$$

Apply the projection theorem for $\mathbf{V} = \mathbf{L} + 2\mathbf{S}$ within an $|LSJM_J\rangle$ manifold:

$$\langle \mathbf{V} \rangle = \frac{\langle \mathbf{V} \cdot \mathbf{J} \rangle}{\langle \mathbf{J}^2 \rangle} \langle \mathbf{J} \rangle \equiv g_J \langle \mathbf{J} \rangle.$$

Expand $\mathbf{V} \cdot \mathbf{J} = (\mathbf{L} + 2\mathbf{S}) \cdot (\mathbf{L} + \mathbf{S})$:

$$\mathbf{V} \cdot \mathbf{J} = \mathbf{L}^2 + \mathbf{L} \cdot \mathbf{S} + 2\mathbf{S} \cdot \mathbf{L} + 2\mathbf{S}^2 = \mathbf{L}^2 + 3\mathbf{L} \cdot \mathbf{S} + 2\mathbf{S}^2.$$

Solve $\mathbf{J}^2 = \mathbf{L}^2 + \mathbf{S}^2 + 2\mathbf{L} \cdot \mathbf{S}$ for $\mathbf{L} \cdot \mathbf{S}$:

$$\mathbf{L} \cdot \mathbf{S} = \frac{1}{2}(\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2).$$

Substitute into $\mathbf{V} \cdot \mathbf{J}$:

$$\mathbf{V} \cdot \mathbf{J} = \mathbf{L}^2 + \frac{3}{2}(\mathbf{J}^2 - \mathbf{L}^2 - \mathbf{S}^2) + 2\mathbf{S}^2.$$

Collect terms:

$$\mathbf{V} \cdot \mathbf{J} = \frac{3}{2}\mathbf{J}^2 - \frac{1}{2}\mathbf{L}^2 + \frac{1}{2}\mathbf{S}^2.$$

Take expectation in $|LSJM_J\rangle$ and divide by $\langle \mathbf{J}^2 \rangle = J(J+1)\hbar^2$:

$$g_J = \frac{\frac{3}{2}J(J+1) - \frac{1}{2}L(L+1) + \frac{1}{2}S(S+1)}{J(J+1)}.$$

Rewrite $3/2 = 1 + 1/2$:

$$g_J = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)}.$$

Energy shift in eigenstate $|LSJM_J\rangle$:

$$\Delta E_Z = g_J \mu_B B M_J, \quad M_J = -J, \dots, J.$$

(c) Zeeman patterns for the two transitions

$1s3s^1S_0 \rightarrow 1s2p^1P_1$ (*singlet*): $S = 0$ in both terms, so $g_J = 1$ for both — **normal Zeeman effect**.

Upper $J = 0$: $M_J = 0$ only, no splitting. Lower $J = 1$: $M_J = -1, 0, +1$, shifts $-\mu_B B, 0, +\mu_B B$.

Write the emission frequency for $u \rightarrow l$:

$$\nu = \nu_0 + (g_u M_u - g_l M_l) \mu_B B / \hbar.$$

Set $g_u = g_l = 1$ and $M_u = 0$:

$$\nu = \nu_0 - M_l \mu_B B / \hbar.$$

Insert $M_l = -1, 0, +1$:

$$\nu = \nu_0 + \mu_B B / \hbar, \quad \nu_0, \quad \nu_0 - \mu_B B / \hbar.$$

Polarisations:

- $\Delta M_J = M_u - M_l = \pm 1$: $\sigma^\pm$ (circular along $\mathbf{B}$; linear $\perp \mathbf{B}$ transverse).
- $\Delta M_J = 0$: π (linear $\parallel \mathbf{B}$; visible only in transverse observation).

$1s3s\ ^3S_1 \rightarrow 1s2p\ ^3P_1$ (triplet): **anomalous Zeeman effect**.

Compute Landé for upper 3S_1 ($L = 0, S = 1, J = 1$):

$$g(^3S_1) = 1 + \frac{1 \cdot 2 + 1 \cdot 2 - 0}{2 \cdot 1 \cdot 2} = 1 + \frac{4}{4} = 2.$$

Compute Landé for lower 3P_1 ($L = 1, S = 1, J = 1$):

$$g(^3P_1) = 1 + \frac{1 \cdot 2 + 1 \cdot 2 - 1 \cdot 2}{2 \cdot 1 \cdot 2} = 1 + \frac{2}{4} = \frac{3}{2}.$$

Sub-level shifts in units of $\mu_B B$: upper $\{-2, 0, +2\}$; lower $\{-3/2, 0, +3/2\}$.

Write the emission frequency:

$$\nu = \nu_0 + (g_u M_u - g_l M_l) \mu_B B / h = \nu_0 + (2M_u - \frac{3}{2}M_l) \mu_B B / h.$$

Apply selection rules $\Delta M = 0, \pm 1$ with $M = 0 \not\rightarrow M = 0$ for $\Delta J = 0$.

Enumerate σ^+ ($M_u - M_l = +1$): $(M_u, M_l) = (+1, 0)$ and $(0, -1)$:

$$(2 \cdot 1 - 0) = +2, \quad (0 - \frac{3}{2}(-1)) = +\frac{3}{2}.$$

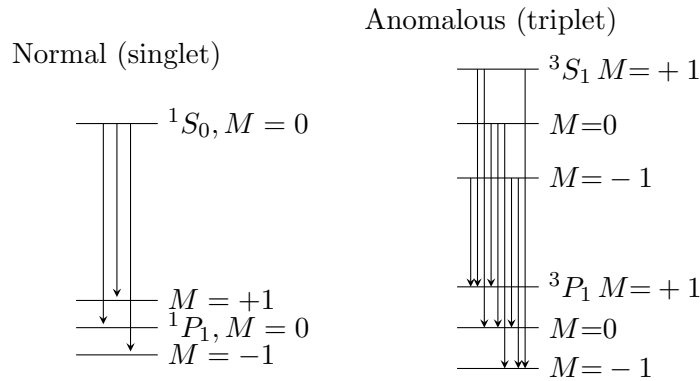
Enumerate σ^- ($M_u - M_l = -1$): $(M_u, M_l) = (-1, 0)$ and $(0, +1)$:

$$(2 \cdot (-1) - 0) = -2, \quad (0 - \frac{3}{2} \cdot 1) = -\frac{3}{2}.$$

Enumerate allowed π ($\Delta M = 0, M \neq 0$): $(M_u, M_l) = (\pm 1, \pm 1)$:

$$(2 - \frac{3}{2}) = +\frac{1}{2}, \quad (-2 + \frac{3}{2}) = -\frac{1}{2}.$$

Total: 4 σ and 2 π components, so 6 triplet lines.



(d) Six lines $\Rightarrow$ orientation; minimum B

Total components: triplet gives $4\sigma + 2\pi = 6$; singlet gives $2\sigma + 1\pi = 3$; total 9. Observing only 6 means all 3 π components are absent. π -light has $\mathbf{E} \parallel \mathbf{B}$ and so radiates only transverse to $\mathbf{B}$; it vanishes in longitudinal observation. Conclude:

Observer is along the magnetic field, $\mathbf{k} \parallel \mathbf{B}$; only $\sigma^\pm$ visible: $4 + 2 = 6$ lines.

Write the pressure-broadened linewidth (FWHM):

$$\Delta\nu = \frac{1}{2\pi\tau_{\text{coll}}}, \quad \tau_{\text{coll}}^{-1} = n\sigma\bar{v}, \quad \sigma = \pi(2r)^2.$$

Compute the number density from the ideal-gas law:

$$n = \frac{P}{k_B T} = \frac{0.1}{1.381 \times 10^{-23} \times 10}.$$

Evaluate:

$$n = 7.24 \times 10^{20} \text{ m}^{-3}.$$

Write the Maxwell–Boltzmann mean speed:

$$\bar{v} = \sqrt{\frac{8k_B T}{\pi m_{\text{He}}}}.$$

Insert $m_{\text{He}} = 4 \times 1.66 \times 10^{-27} \text{ kg} = 6.64 \times 10^{-27} \text{ kg}$ and $T = 10 \text{ K}$:

$$\bar{v} = \sqrt{\frac{8 \cdot 1.381 \times 10^{-23} \cdot 10}{\pi \cdot 6.64 \times 10^{-27}}}.$$

Simplify the radicand:

$$\bar{v} = \sqrt{5.30 \times 10^4 \text{ m}^2/\text{s}^2}.$$

Evaluate:

$$\bar{v} = 230 \text{ m/s}.$$

Compute the hard-sphere cross-section with $r = 0.1 \text{ nm}$:

$$\sigma = \pi(2r)^2 = \pi(0.2 \times 10^{-9} \text{ m})^2 = 1.26 \times 10^{-19} \text{ m}^2.$$

Compute the collision rate:

$$\tau_{\text{coll}}^{-1} = n\sigma\bar{v} = 7.24 \times 10^{20} \cdot 1.26 \times 10^{-19} \cdot 230.$$

Evaluate:

$$\tau_{\text{coll}}^{-1} = 2.10 \times 10^4 \text{ s}^{-1}.$$

Divide by 2π to get the linewidth:

$$\Delta\nu = \frac{2.10 \times 10^4}{2\pi} = 3.34 \times 10^3 \text{ Hz}.$$

List σ -shifts about each line centre (units $\mu_B B/h$): singlet $\{-1, +1\}$ and triplet $\{-2, -\frac{3}{2}, +\frac{3}{2}, +2\}$.

Identify the smallest adjacent gap (between triplet shifts -2 and $-3/2$, or between $+3/2$ and $+2$):

$$\Delta\nu_{\text{min}} = \frac{1}{2} \mu_B B/h.$$

Equate $\Delta\nu_{\text{min}} = \Delta\nu$ and solve for B :

$$B_{\text{min}} = \frac{2h \Delta\nu}{\mu_B}.$$

Insert numbers:

$$B_{\text{min}} = \frac{2 \cdot 6.626 \times 10^{-34} \cdot 3.34 \times 10^3}{9.274 \times 10^{-24}}.$$

Evaluate the numerator:

$$2 \cdot 6.626 \times 10^{-34} \cdot 3.34 \times 10^3 = 4.43 \times 10^{-30} \text{ J s Hz}.$$

Divide by μ_B :

$$B_{\text{min}} \approx 4.8 \times 10^{-7} \text{ T} \sim 0.5 \mu\text{T}.$$

(e) Positronium ground configuration

Electron and positron are both spin-1/2. Couple $1/2 \otimes 1/2 = 0 \oplus 1$:

$$S = 0 \text{ (para-positronium, } {}^1S_0\text{), } \quad S = 1 \text{ (ortho-positronium, } {}^3S_1\text{).}$$

Apply $\mathbf{B} = B\hat{\mathbf{z}}$. Electron moment $\boldsymbol{\mu}_{e^-} = -g_s\mu_B\mathbf{S}_{e^-}/\hbar$; positron has opposite-sign g -factor, so $\boldsymbol{\mu}_{e^+} = +g_s\mu_B\mathbf{S}_{e^+}/\hbar$.

Sum the moments:

$$\boldsymbol{\mu}_{e^-} + \boldsymbol{\mu}_{e^+} = -\frac{g_s\mu_B}{\hbar}(\mathbf{S}_{e^-} - \mathbf{S}_{e^+}).$$

Form the Zeeman Hamiltonian $H_Z = -(\boldsymbol{\mu}_{e^-} + \boldsymbol{\mu}_{e^+}) \cdot \mathbf{B}$:

$$H_Z = \frac{g_s\mu_B B}{\hbar}(S_z^{e^-} - S_z^{e^+}).$$

Write the singlet and triplet $M = 0$ kets:

$$|1, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle), \quad |0, 0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle).$$

Act with $S_z^{e^-} - S_z^{e^+}$ on each product ket:

$$(S_z^{e^-} - S_z^{e^+})|\uparrow\downarrow\rangle = \frac{\hbar}{2}|\uparrow\downarrow\rangle - (-\frac{\hbar}{2})|\uparrow\downarrow\rangle = \hbar|\uparrow\downarrow\rangle,$$

$$(S_z^{e^-} - S_z^{e^+})|\downarrow\uparrow\rangle = -\frac{\hbar}{2}|\downarrow\uparrow\rangle - \frac{\hbar}{2}|\downarrow\uparrow\rangle = -\hbar|\downarrow\uparrow\rangle.$$

Apply to $|1, 0\rangle$:

$$(S_z^{e^-} - S_z^{e^+})|1, 0\rangle = \frac{\hbar}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) = \hbar|0, 0\rangle.$$

Take the inner product with $\langle 0, 0|$:

$$\langle 0, 0| (S_z^{e^-} - S_z^{e^+}) |1, 0\rangle = \hbar.$$

Multiply by the prefactor of H_Z :

$$\langle 0, 0| H_Z |1, 0\rangle = g_s\mu_B B.$$

The triplet $M = \pm 1$ states $|\uparrow\uparrow\rangle, |\downarrow\downarrow\rangle$ satisfy $(S_z^{e^-} - S_z^{e^+})|\pm\pm\rangle = 0$, so they are unshifted to first order.

Effect: the field **mixes the singlet** $|0, 0\rangle$ **with the triplet** $|1, 0\rangle$ rather than producing the usual linear Zeeman splitting. The two extreme triplet sublevels $M = \pm 1$ are unshifted to leading order; the two $M = 0$ states repel each other quadratically in B (an avoided crossing). This is the textbook Breit–Rabi ortho–para mixing in positronium hyperfine spectroscopy.

Question 3 — Rabi oscillations and three-level dynamics

(a) Meaning of Ω_R

Semiclassical dipole interaction: $H_{\text{int}} = -\mathbf{d} \cdot \mathbf{E}_0 \cos \omega t$. Rabi frequency:

$$\Omega_R = \frac{|\langle 2| \mathbf{d} \cdot \hat{\mathbf{e}} |1\rangle| E_0}{\hbar}$$

Ω_R is the **on-resonance Rabi frequency** = oscillation frequency of population between $|1\rangle$ and $|2\rangle$ on resonance, set by the dipole matrix element times the field amplitude (in units of $\hbar$). Period $2\pi/\Omega_R$ corresponds to a 2π pulse.

(b) Rotating wave approximation; weak-light solution

Write $\cos \omega t = (e^{i\omega t} + e^{-i\omega t})/2$ in $\dot{c}_2$:

$$\dot{c}_2 = -i\frac{\Omega_R}{2} [e^{i(\omega_0+\omega)t} + e^{i(\omega_0-\omega)t}] c_1.$$

RWA: drop the rapidly oscillating term $e^{i(\omega_0+\omega)t}$:

$$\dot{c}_2 = -i\frac{\Omega_R}{2} e^{-i\delta\omega t} c_1, \quad \delta\omega \equiv \omega - \omega_0.$$

Similarly for $\dot{c}_1$:

$$\dot{c}_1 = -i\frac{\Omega_R}{2} e^{i\delta\omega t} c_2.$$

Differentiate the $\dot{c}_2$ equation:

$$\ddot{c}_2 = -i\frac{\Omega_R}{2} [-i\delta\omega e^{-i\delta\omega t} c_1 + e^{-i\delta\omega t} \dot{c}_1].$$

Identify $-i(\Omega_R/2)e^{-i\delta\omega t} c_1 = \dot{c}_2$:

$$\ddot{c}_2 = -i\delta\omega \dot{c}_2 - i\frac{\Omega_R}{2} e^{-i\delta\omega t} \dot{c}_1.$$

Substitute $\dot{c}_1 = -i(\Omega_R/2)e^{i\delta\omega t} c_2$:

$$\ddot{c}_2 = -i\delta\omega \dot{c}_2 + (-i)^2 \frac{\Omega_R^2}{4} e^{-i\delta\omega t} e^{i\delta\omega t} c_2.$$

Cancel the exponentials and use $(-i)^2 = -1$:

$$\ddot{c}_2 = -i\delta\omega \dot{c}_2 - \frac{\Omega_R^2}{4} c_2.$$

Rearrange to standard form:

$$\ddot{c}_2 + i\delta\omega \dot{c}_2 + \frac{\Omega_R^2}{4} c_2 = 0.$$

Try $c_2 = Ae^{i\lambda t}$, which gives $\ddot{c}_2 = -\lambda^2 c_2$ and $\dot{c}_2 = i\lambda c_2$:

$$-\lambda^2 + i\delta\omega(i\lambda) + \frac{\Omega_R^2}{4} = 0.$$

Simplify:

$$-\lambda^2 - \delta\omega \lambda + \frac{\Omega_R^2}{4} = 0.$$

Multiply by -1 :

$$\lambda^2 + \delta\omega \lambda - \frac{\Omega_R^2}{4} = 0.$$

Solve the quadratic:

$$\lambda = \frac{1}{2} [-\delta\omega \pm \sqrt{\delta\omega^2 + \Omega_R^2}].$$

Define the generalised Rabi frequency:

$$\alpha \equiv \sqrt{\Omega_R^2 + \delta\omega^2}.$$

Write the general solution:

$$c_2(t) = e^{-i\delta\omega t/2} (A_+ e^{i\alpha t/2} + A_- e^{-i\alpha t/2}).$$

Apply $c_2(0) = 0$, giving $A_+ + A_- = 0$, hence $A_- = -A_+ \equiv -A$:

$$c_2(t) = A e^{-i\delta\omega t/2} (e^{i\alpha t/2} - e^{-i\alpha t/2}).$$

Use $e^{ix} - e^{-ix} = 2i \sin x$:

$$c_2(t) = 2iA e^{-i\delta\omega t/2} \sin(\alpha t/2).$$

Differentiate and evaluate at $t = 0$:

$$\dot{c}_2(0) = 2iA \cdot (\alpha/2) = iA\alpha.$$

Match to the EOM at $t = 0$ with $c_1(0) = 1$: $\dot{c}_2(0) = -i\Omega_R/2$:

$$iA\alpha = -i\Omega_R/2 \implies A = -\frac{\Omega_R}{2\alpha}.$$

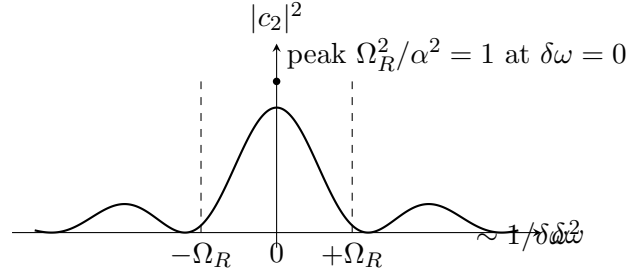
Substitute back:

$$c_2(t) = -\frac{i\Omega_R}{\alpha} e^{-i\delta\omega t/2} \sin(\alpha t/2).$$

Take modulus squared:

$$|c_2(t)|^2 = \frac{\Omega_R^2}{\alpha^2} \sin^2(\alpha t/2), \quad \alpha = \sqrt{\Omega_R^2 + (\omega - \omega_0)^2}.$$

(c) Sketch vs. detuning



Central peak height 1 (full transfer at resonance, fixed t chosen so $\sin^2(\Omega_R t/2) = 1$); side lobes decay $\sim 1/\delta\omega^2$, half-width $\sim \Omega_R$.

(d) Three-level: $c_2(t)$ to leading order

Take initial conditions $c_1(0) = 1$, $c_2(0) = c_3(0) = 0$; in weak light keep $c_1 \approx 1$ and $c_3 \approx 0$:

$$\dot{c}_2 \approx -i\Omega_R \cos \omega t e^{i\omega_0 t}.$$

Expand $\cos \omega t$:

$$\dot{c}_2 \approx -i\frac{\Omega_R}{2} [e^{i(\omega_0+\omega)t} + e^{i(\omega_0-\omega)t}].$$

Apply RWA (drop $e^{i(\omega_0+\omega)t}$, keep $e^{-i\delta\omega t}$):

$$\dot{c}_2 \approx -i\frac{\Omega_R}{2} e^{-i\delta\omega t}.$$

Integrate from 0 to t with $c_2(0) = 0$:

$$c_2(t) = -i\frac{\Omega_R}{2} \int_0^t e^{-i\delta\omega t'} dt'.$$

Evaluate the integral:

$$c_2(t) = -i\frac{\Omega_R}{2} \cdot \frac{1 - e^{-i\delta\omega t}}{i\delta\omega}.$$

Cancel the i 's:

$$c_2(t) \approx -\frac{\Omega_R}{2\delta\omega} (1 - e^{-i\delta\omega t}).$$

Amplitude $\sim \Omega_R/\delta\omega \ll 1$ when $\delta\omega \gg \Omega_R$: the off-resonant intermediate level is only weakly admixed.

(e) Effective two-photon Rabi flopping for c_3

Define single-photon detunings $\delta\omega \equiv \omega - \omega_0$ and $\omega - \phi_0$; the two-photon detuning is $\Delta\omega = \omega_0 + \phi_0 - 2\omega$. Near two-photon resonance $\Delta\omega \rightarrow 0$, so $\delta\omega \approx \phi_0 - \omega$; both intermediate detunings are large compared to Ω_R, Φ_R .

Expand $\cos \omega t$ in the $\dot{c}_2$ equation:

$$\dot{c}_2 = -i\frac{\Omega_R}{2} [e^{i(\omega_0+\omega)t} + e^{i(\omega_0-\omega)t}] c_1 - i\frac{\Phi_R}{2} [e^{-i(\phi_0-\omega)t} + e^{-i(\phi_0+\omega)t}] c_3.$$

Drop the fast-rotating terms $e^{\pm i(\omega_0+\omega)t}, e^{-i(\phi_0+\omega)t}$:

$$\dot{c}_2 = -i\frac{\Omega_R}{2} e^{-i\delta\omega t} c_1 - i\frac{\Phi_R}{2} e^{-i(\phi_0-\omega)t} c_3.$$

Adiabatically eliminate c_2 : with c_1, c_3 slow, c_2 follows the driving. Take the ansatz $c_2 = a e^{-i\delta\omega t} c_1 + b e^{-i(\phi_0-\omega)t} c_3$ and demand $\dot{c}_2$ match:

$$-i\delta\omega a e^{-i\delta\omega t} c_1 - i(\phi_0 - \omega) b e^{-i(\phi_0-\omega)t} c_3 = -i\frac{\Omega_R}{2} e^{-i\delta\omega t} c_1 - i\frac{\Phi_R}{2} e^{-i(\phi_0-\omega)t} c_3.$$

Match coefficients of $e^{-i\delta\omega t} c_1$:

$$a = \frac{\Omega_R}{2\delta\omega}.$$

Match coefficients of $e^{-i(\phi_0-\omega)t} c_3$:

$$b = \frac{\Phi_R}{2(\phi_0 - \omega)}.$$

Assemble c_2 :

$$c_2 \approx \frac{\Omega_R}{2\delta\omega} e^{-i\delta\omega t} c_1 + \frac{\Phi_R}{2(\phi_0 - \omega)} e^{-i(\phi_0-\omega)t} c_3.$$

Take the RWA $\dot{c}_3$ equation:

$$\dot{c}_3 = -i\frac{\Phi_R}{2} e^{i(\phi_0-\omega)t} c_2.$$

Substitute c_2 :

$$\dot{c}_3 = -i\frac{\Phi_R}{2} \left[\frac{\Omega_R}{2\delta\omega} e^{i(\phi_0-\omega-\delta\omega)t} c_1 + \frac{\Phi_R}{2(\phi_0-\omega)} c_3 \right].$$

Combine exponents using $(\phi_0 - \omega) - \delta\omega = \omega_0 + \phi_0 - 2\omega = \Delta\omega$:

$$\dot{c}_3 = -i\frac{\Omega_R\Phi_R}{4\delta\omega} e^{i\Delta\omega t} c_1 - i\frac{\Phi_R^2}{4(\phi_0-\omega)} c_3.$$

Identify the AC-Stark light shift on level 3:

$$\delta_3 = \frac{\Phi_R^2}{4(\phi_0 - \omega)}.$$

Repeat the elimination for $\dot{c}_1 = -i(\Omega_R/2)e^{i\delta\omega t} c_2$:

$$\dot{c}_1 = -i\frac{\Omega_R^2}{4\delta\omega} c_1 - i\frac{\Omega_R\Phi_R}{4(\phi_0-\omega)} e^{-i\Delta\omega t} c_3.$$

Identify the AC-Stark shift on level 1:

$$\delta_1 = \frac{\Omega_R^2}{4\delta\omega}.$$

Near two-photon resonance set $\delta\omega \approx \phi_0 - \omega$ in the off-diagonal terms and define the effective two-photon Rabi frequency:

$$\Omega_{\text{eff}} \equiv \frac{\Omega_R\Phi_R}{2\delta\omega}.$$

The reduced two-level system reads:

$$\dot{c}_1 = -i\delta_1 c_1 - i\frac{\Omega_{\text{eff}}}{2} e^{-i\Delta\omega t} c_3,$$

$$\dot{c}_3 = -i\delta_3 c_3 - i\frac{\Omega_{\text{eff}}}{2} e^{i\Delta\omega t} c_1.$$

Absorb the diagonal shifts into a modified detuning $\Delta\omega' = \Delta\omega + \delta_3 - \delta_1$, then apply the result of part (b):

$$|c_3(t)|^2 = \frac{\Omega_{\text{eff}}^2}{\beta^2} \sin^2(\beta t/2), \quad \Omega_{\text{eff}} = \frac{\Omega_R \Phi_R}{2\delta\omega}, \quad \beta = \sqrt{\Omega_{\text{eff}}^2 + (\Delta\omega + \delta_3 - \delta_1)^2}.$$

(Often the AC-Stark shifts are absorbed into a redefined two-photon detuning, leaving $\beta = \sqrt{\Omega_{\text{eff}}^2 + \Delta\omega^2}$.)

Question 4 — Einstein coefficients and astrophysical lasers

(a) Einstein A , B_{ul} , B_{lu}

Three processes between levels u (energy E_u) and l (E_l), $h\nu_0 = E_u - E_l$, in radiation field of spectral energy density $\rho(\nu)$:

- **Spontaneous emission**, A_{ul} : rate per atom A_{ul} . Power emitted: $N_u A_{ul} h\nu_0$.
- **Stimulated emission**, B_{ul} : rate per atom $B_{ul}\rho(\nu_0)$. Power emitted: $N_u B_{ul}\rho(\nu_0) h\nu_0$.
- **Absorption**, B_{lu} : rate per atom $B_{lu}\rho(\nu_0)$. Power absorbed: $N_l B_{lu}\rho(\nu_0) h\nu_0$.

Detailed balance (thermal, Planck): $g_l B_{lu} = g_u B_{ul}$ and $A_{ul}/B_{ul} = 8\pi h\nu_0^3/c^3$.

(b) $I_S \propto \nu_0^3$ for radiative-only upper level

Write the gain cross section at line centre (Lorentzian, $\nu_L \approx \nu_0$):

$$\sigma_{ul}(\nu_0) = \frac{\lambda_0^2}{8\pi} A_{ul} \phi(\nu_0), \quad \phi(\nu_0) = \frac{2}{\pi\Delta\nu}.$$

Insert $\phi(\nu_0)$:

$$\sigma_{ul}(\nu_0) = \frac{\lambda_0^2 A_{ul}}{4\pi^2 \Delta\nu}.$$

Upper level radiatively dominated: $\tau_u = 1/A_{ul}$, equivalently $\tau_u A_{ul} = 1$.

Factor $(g_u/g_l)\tau_l$ in the bracket of I_S :

$$\tau_u + \frac{g_u}{g_l} \tau_l - \frac{g_u}{g_l} \tau_u \tau_l A_{ul} = \tau_u + \frac{g_u}{g_l} \tau_l (1 - \tau_u A_{ul}).$$

Apply $\tau_u A_{ul} = 1$:

$$\tau_u + \frac{g_u}{g_l} \tau_l (1 - 1) = \tau_u = \frac{1}{A_{ul}}.$$

Insert into the saturation formula:

$$I_S = \frac{h\nu_L}{\sigma_{ul}} (1/A_{ul})^{-1} = \frac{h\nu_L A_{ul}}{\sigma_{ul}}.$$

Substitute $\sigma_{ul} = \lambda_0^2 A_{ul}/(4\pi^2 \Delta\nu)$:

$$I_S = h\nu_L A_{ul} \cdot \frac{4\pi^2 \Delta\nu}{\lambda_0^2 A_{ul}}.$$

Cancel A_{ul} :

$$I_S = \frac{4\pi^2 h \nu_L \Delta \nu}{\lambda_0^2}.$$

Use $\lambda_0 = c/\nu_0$, so $\lambda_0^2 = c^2/\nu_0^2$:

$$I_S = \frac{4\pi^2 h \nu_L \nu_0^2 \Delta \nu}{c^2}.$$

Set $\nu_L \approx \nu_0$:

$$I_S = \frac{4\pi^2 h \nu_0^3 \Delta \nu}{c^2} \propto \nu_0^3.$$

(c) Spontaneous-emission photon rate from the seed region

Compute the total spontaneous rate from the cylinder (volume $\pi d^2 \ell/4$, density N_u):

$$\dot{N}_\gamma^{\text{spon}} = \frac{\pi d^2}{4} \ell N_u A_{ul}.$$

Use $N_u \approx N^*$ for a large initial inversion ($N_l \ll N_u$):

$$\dot{N}_\gamma^{\text{spon}} = \frac{\pi d^2}{4} N^* \ell A_{ul}.$$

Insert the gain-length condition $\sigma_{ul} N^* \ell = 1$ as $N^* \ell = 1/\sigma_{ul}$:

$$\dot{N}_\gamma^{\text{spon}} = \frac{\pi d^2}{4} \frac{A_{ul}}{\sigma_{ul}}.$$

From part (b), $I_S = h \nu_L A_{ul} / \sigma_{ul}$, hence $A_{ul} / \sigma_{ul} = I_S / (h \nu_0)$:

$$\dot{N}_\gamma^{\text{spon}} = \frac{\pi d^2 I_S}{4 h \nu_0}.$$

(d) Minimum length to reach saturation

In the unsaturated regime N^* is constant. Write the gain equation:

$$\frac{dI}{dz} = \sigma_{ul} N^* I.$$

Use $\sigma_{ul} N^* \ell = 1$:

$$\frac{dI}{dz} = \frac{I}{\ell}.$$

Integrate with seed intensity I_0 at $z = 0$:

$$I(z) = I_0 e^{z/\ell}.$$

Estimate the seed intensity: spontaneous power forward = $\eta \cdot h \nu_0 A_{ul} N_u \cdot (\pi d^2 \ell/4)$ over beam area $\pi d^2/4$:

$$I_0 = \eta h \nu_0 A_{ul} N_u \ell.$$

Saturation occurs when $I(L_S) = I_S$:

$$I_S = I_0 e^{L_S/\ell}.$$

Take the logarithm:

$$\frac{L_S}{\ell} = \ln\left(\frac{I_S}{I_0}\right).$$

Substitute I_0 and split the log:

$$\frac{L_S}{\ell} = \ln(1/\eta) + \ln\left(\frac{I_S}{h\nu_0 A_{ul} N_u \ell}\right).$$

Use $I_S = h\nu_0 A_{ul}/\sigma_{ul}$ (part b) in the second log:

$$\frac{I_S}{h\nu_0 A_{ul} N_u \ell} = \frac{1}{\sigma_{ul} N_u \ell}.$$

Apply $\sigma_{ul} N^* \ell = 1$ with $N^* \approx N_u$:

$$\frac{1}{\sigma_{ul} N_u \ell} = 1.$$

The second logarithm vanishes:

$$\frac{L_S}{\ell} = \ln(1/\eta).$$

Use $\ell = 1/(\sigma_{ul} N_u) = I_S/(h\nu_0 A_{ul} N_u)$:

$$L_S \simeq \frac{I_S}{h\nu_0 A_{ul} N_u} \ln(1/\eta).$$

(e) Estimate of η

Photons are emitted isotropically from the seed region. The fraction reaching the far end is the geometric ratio of its solid angle:

$$\eta \sim \frac{\Delta\Omega}{4\pi} = \frac{\pi d^2/4}{4\pi L^2}.$$

Simplify:

$$\eta \sim \frac{d^2}{16 L^2}.$$

For $d \ll L$ this is small but enters L_S only logarithmically: $\ln(1/\eta) \sim 2 \ln(4L/d)$.